

SANDADINI BANDARA

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Electronics Engineer · Embedded Systems · Robotics & IoT



PROFILE SUMMARY

Electronics and Information Technology graduate from the University of Colombo with hands-on experience in embedded systems, PCB design, robotics, automation, and IoT systems. Proven track record of building and testing hardware projects involving microcontrollers, wireless communication, sensor integration, and real-time control systems. Proficient in PCB schematic design, embedded C and Python development, hardware debugging, and actuator interfacing. Strong practical exposure to robotics competitions, industrial automation workflows, and multidisciplinary engineering projects.

EDUCATION

University of Colombo — BSc (Hons) in Electronics and Information Technology

2022 – 2026

Relevant Coursework: Data Structures & Algorithms, Software Engineering, Embedded Systems, Robotics & Automation, Power Electronics, Analog & Digital Electronics, Circuit Design & Simulation, Internet Technologies, Database Systems, Industrial Automation, Data Acquisition

PROFESSIONAL EXPERIENCE

Sri Lanka Telecom — Electronic & IT Intern

2026

- Designed and debugged schematic diagrams and multi-layer PCB layouts using EasyEDA, improving prototype reliability and reducing hardware troubleshooting time.
- Programmed embedded microcontroller systems with GSM integration to enable real-time remote alert and notification functionality.
- Performed hardware testing, signal verification, and fault diagnosis using oscilloscopes, multimeters, and logic analyzers across embedded and telecom systems.
- Supported hardware-software integration activities for embedded devices deployed in telecom and field environments.
- Collaborated with cross-functional technical teams on embedded system validation, wiring verification, and fault-resolution tasks under real-world operational conditions.

PROJECTS

Underground Wire Detecting Robot

Developed a mobile robot to detect underground wires using an ATmega328p microcontroller and magnetometer sensor. Programmed in C and Python, integrating embedded electronics for real-time detection and data visualisation.

Repository: [Underground Wire Detector Robot](#)

Underwater Explorer

Designed and built an underwater exploration robot powered by Raspberry Pi 4. Implemented Python-based control logic for motor actuation, live video streaming, underwater photography, and real-time temperature monitoring.

Repository: [Underwater Rover](#)

Voice-Activated Emergency Alert System

Engineered a voice-controlled alert system using Arduino Nano 33 BLE and a GSM SIM900 module. Integrated a custom Edge Impulse machine learning model for voice recognition, enabling real-time SMS alerts in emergency situations.

CityLink - Smart Public Transport Booking Application

Developed a React Native mobile application for centralised seat booking, timetable viewing, and real-time bus tracking. Built separate passenger and conductor interfaces with QR code scanning. Backend implemented in Node.js and MySQL; UI designed in Figma.

Repository: [CityLink](#)

Line Following Robot

Designed and implemented an autonomous line-following robot using Arduino, combining electronic circuit design with Python scripting for performance monitoring and data analysis.

Attendance Management System

Developed a web-based attendance tracking platform using Java, PHP, and CodeIgniter. Automated record-keeping by enabling lecturers to scan student IDs via smartphone, with instant database updates and real-time status notifications.

Special Selection Mobile Application

Created an Android application (Java, Android Studio) to automate student selection workflows. The app generates ranked selection lists from uploaded student preferences and exports results in CSV format.

Repository: [SpecializedU](#)

Glowing Road-Marking Paint - Case Study

Conducted a research-based case study on nano-silicon glass beads in luminescent paint for road safety. Proposed sustainable material solutions to improve nighttime visibility and reduce accident risk.

TECHNICAL SKILLS

Programming Languages: C, Python, Java, C++, JavaScript, PHP, HTML/CSS

Embedded Platforms: ESP32, ATmega328p, Arduino, Raspberry Pi, STM8S

Electronics & Test Equipment: Oscilloscope, Logic Analyser, Multimeter, Power Supplies

Software & Design Tools: EasyEDA, Altium Designer, Fusion 360, Edge Impulse, Android Studio, Linux

Automation & PLC: Ladder Logic Programming, WinProLadder

Web & Mobile Development: React Native, Node.js, MySQL, Java (Android), PHP (CodeIgniter)

COMPETITIONS & ACHIEVEMENTS

Participant - SLIIT Robofest Robotics Competition	2025
Participant - Perabots Robotics Competition	2025
Captain - Women's Chess Team, University of Colombo	2025
Member - Sri Lankan Team, Asian University Chess Championship	2022
SLUSA Colours Award for Chess	2022
Best Female Chess Player, University of Colombo	2024
Deputy Head Prefect	2020
President - Aviation Club	2020

REFERENCES

Dr. Deshitha Wickramaratna

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